

PrimeSim™ Analog Design Environment Integration User Guide

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About This User Guide

This User Guide supports the Synopsys PrimeSim™ Continuum integration to the Cadence® Virtuoso® Analog Design Environment (ADE). The content of this user guide provides information regarding the PrimeSim Continuum integration and analysis/simulation setup. For information on the Cadence tool, refer to the *Virtuoso® Analog Design Environment User Guide* documentation, available through that licensed tool.

All references to the Cadence® Virtuoso® ADE platform are noted in this user guide as the “Environment.” The graphical interface for the PrimeSim Continuum integration displayed in this manual is used with permission and is based on the version 23.10 of the native tool.

This preface includes the following sections:

- [New in This Release](#)
- [Related Products, Publications, and Trademarks](#)
- [Conventions](#)
- [Customer Support](#)
- [Statement on Inclusivity and Diversity](#)

New in This Release

Information about new features, enhancements, and changes, known limitations, and resolved Synopsys Technical Action Requests (STARs) is available in the PrimeSim Continuum Release Notes on the SolvNetPlus site.

Related Products, Publications, and Trademarks

For additional information about the PrimeSim Continuum simulators, see the documentation on the Synopsys SolvNetPlus support site at the following address:

<https://solvnetplus.synopsys.com>

You might also want to see the documentation for the following related Synopsys products:

- The documentation set installed with the PrimeSim Continuum software distribution.
- The Release Notes for the PrimeSim Continuum, which are published with each release and service pack, and are available on SolvNetPlus (see [Accessing SolvNetPlus](#)).

Contents

- Documentation on the Web, that is the PrimeSim Continuum online help and PDF documents now available on SolvNetPlus (see [Accessing SolvNetPlus](#))
- The Cadence™ *Virtuoso® Analog Design Environment User Guide* documentation is available through that licensed tool.

Conventions

The following conventions are used in Synopsys documentation.

Convention	Description
<code>Courier</code>	Indicates syntax, such as <code>write_file</code> .
<i>Courier italic</i>	Indicates a user-defined value in syntax, such as <code>write_file design_list</code>
Courier bold	Indicates user input—text you type verbatim—in examples, such as prompt> write_file top
Purple	• Within an example, indicates information of special interest. • Within a command-syntax section, indicates a default, such as <code>include_enclosing = true false</code>
[]	Denotes optional arguments in syntax, such as <code>write_file [-format fmt]</code>
...	Indicates that arguments can be repeated as many times as needed, such as <code>pin1 pin2 ... pinN</code> .
	Indicates a choice among alternatives, such as <code>low medium high</code>
\	Indicates a continuation of a command line.
/	Indicates levels of directory structure.
Bold	Indicates a graphical user interface (GUI) element that has an action associated with it.
Edit > Copy	Indicates a path to a menu command, such as opening the Edit menu and choosing Copy .
Ctrl+C	Indicates a keyboard combination, such as holding down the Ctrl key and pressing C.

Customer Support

Customer support is available through SolvNetPlus.

Accessing SolvNetPlus

The SolvNetPlus site includes a knowledge base of technical articles and answers to frequently asked questions about Synopsys tools. The SolvNetPlus site also gives you access to a wide range of Synopsys online services including software downloads, documentation, and technical support.

To access the SolvNetPlus site, go to the following address:

<https://solvnetplus.synopsys.com>

If prompted, enter your user name and password. If you do not have a Synopsys user name and password, follow the instructions to sign up for an account.

If you need help using the SolvNetPlus site, click REGISTRATION HELP in the top-right menu bar.

Contacting Customer Support

To contact Customer Support, go to <https://solvnetplus.synopsys.com>.

Statement on Inclusivity and Diversity

Synopsys is committed to creating an inclusive environment where every employee, customer, and partner feels welcomed. We are reviewing and removing exclusionary language from our products and supporting customer-facing collateral. Our effort also includes internal initiatives to remove biased language from our engineering and working environment, including terms that are embedded in our software and IPs. At the same time, we are working to ensure that our web content and software applications are usable to people of varying abilities. You may still find examples of non-inclusive language in our software or documentation as our IPs implement industry-standard specifications that are currently under review to remove exclusionary language.

1

Overview of the PrimeSim ADE Integration

This chapter provides an introduction to the PrimeSim ADE Integration flow, capabilities, and features.

This chapter covers the following topics:

- [Capabilities of PrimeSim ADE Integration](#)
- [Supported Features in PrimeSim SPICE-ADE Integration](#)

Capabilities of PrimeSim ADE Integration

The PrimeSim ADE Integration provides the following capabilities:

1. Analyses setup
 - a. Supports 90% of analyses features in ADE-native choosing Analyses form. Commonly used linear and RF analyses features are fully supported.
2. Save and load states/maestro views
3. Use ADE-native features for best usability
 - a. ADE netlisting procedure
 - b. Direct Plot Form
 - c. VIVA
 - d. Corner/Parameter/Variable setup and sweeping
 - e. Job policy setup
4. Save and run batch scripting Open Command Environment for Analysis (OCEAN)
5. Incorporate PrimeSim SPICE simulator features

a. MTS

b. PrimeSim SPICE Monte Carlo

PrimeSim SPICE Reliability Analyses (TMI/Aging, MOSRA)

6. Support `spectre-format` and `hspice-format` netlist flows

Note:

`Spectre-format` netlist flow (and format) is referred to as third-party throughout this manual.

Supported Features in PrimeSim SPICE-ADE Integration

The following features are supported in PrimeSim SPICE-ADE Integration.

Table 1 Supported Features in PrimeSim SPICE-ADE Integration

Features	Sub Items
Analyses Setup	<code>tran</code> , <code>tran-noise</code> <code>ac</code> , <code>dc</code> , <code>noise</code> , <code>xf</code> , <code>sp</code> , <code>stb</code> , <code>sens</code> <code>pss</code> , <code>pac</code> , <code>pnoise</code> , <code>pxf</code> , <code>psp</code> , <code>pstb</code> <code>hb</code> , <code>hbac</code> , <code>hbnoise</code> , <code>hbxf</code> , <code>hbsp</code> , <code>hbstb</code>
Outputs Setup	Add outputs by “Direct Plot Form” All other methods to add outputs in ADE Import / Export outputs expressions to file Save options Format: <code>psf</code> , <code>fsdb</code>
Outputs Evaluation	All of ADE outputs evaluation features Plotting by VIVA Plotting by WaveView
ADE Versions	ADE - L ADE Explorer
Save / load Maestro	ADE-native “save / load states” ADE-native “save / load maestro views”
Simulation Setup	ADE-native “Model Libraries” form ADE-native “Simulation Files” form ADE-native “Stimuli” form ADE-native Variables setup and sweeping ADE-native Parameters setup and sweeping ADE-native Corners setup and sweeping

Table 1 Supported Features in PrimeSim SPICE-ADE Integration (Continued)

Features	Sub Items
High Performance Options	PrimeSim high performance options
Distributed Mode (non-MC simulations)	ADE-native Job Policy Setup ADE-native Job Monitor
Design	Schematic View / Configure View
PrimeSim SPICE Monte - Carlo	Setup and Post Processing for Linear Analyses Job Policy “Command Line” and “Max. Jobs” PrimeSim SPICE Distributed Mode Monitor
PrimeSim SPICE TMI Aging Analyses	PrimeSim SPICE Reliability Setup (TMI / Aging, MOSRA) PrimeSim SPICE Aging Result Post Processing
PrimeSim MTS	Setup and Post Processing

2

Installation

This chapter provides the installation details for the PrimeSim ADE Integration.

This chapter covers the following topics:

- [Installing the PrimeSim ADE Integration](#)
- [Installing the PrimeSim ADE Integration Interface](#)

Installing the PrimeSim ADE Integration

Install the PrimeSim simulator using the standard installation procedure. For example, if your installation directory is `$PRIMESIM_HOME`, the **PrimeSim ADE Integration** interface directory is as follows:

For PrimeSim_SPICE and PrimeSim_Pro:

```
$PRIMESIM_HOME
/interface
/spiceADE
/snpsSimADE.ile
.....
```

For PrimeSim_XA:

```
$XA_HOME
/ade
/PrimeSimXAADE
/snpsSimADE.ile
.....
```

For PrimeSim_HSPICE:

```
$HSPICE_HOME
/hspice
/interface
/snpsSimADE.ile
.....
```

Note:

`snpsSimADE.ile` is a script for **ADE** interface.

Installing the PrimeSim ADE Integration Interface

Install the PrimeSim ADE Integration interface locally without administrator privileges, perform the following steps:

1. Set the Virtuoso environment variables.

a. Set `CDSHOME` or `CDS_ROOT` to the correct Virtuoso path.

```
setenv CDSHOME <path_to_CDS_HOME>
```

b. Set `CDS_LOAD_ENV` to `CSF` to locate the `.cdsenv` files for PrimeSim.

```
setenv CDS_LOAD_ENV CSF
```

The Cadence initialization files are named as `.cdsenv` and `.cdsinit`.

c. Set the `CDS_XVFB_PATH` variable to installation path of the X server.

Use the `which xvfb` command to find the path of `xvfb`. Then use the following path to set the variable.

For example, if `xvfb` is installed in `/usr/local/bin`, then:

```
setenv CDS_XVFB_PATH /usr/local/bin/
```

2. Edit the local `.cdsinit` file.

a. Set the following mandatory settings:

i. Load the script for the **ADE** interface before using any PrimeSim variables in the `.cdsinit` file.

```
load("<absolute_path_to_PRIMESIM_HOME>/interface/spiceADE/snpsSimADE.ile")
```

ii. Set PrimeSim as the default simulator using one of the following examples:

▪ If the default is PrimeSim_SPICE, use the following command:

```
envSetVal("asimenv.startup" "simulator" 'string  
"PrimeSim_SPICE")
```

▪ If the default is PrimeSim_XA, use the following command:

```
envSetVal("asimenv.startup" "simulator" 'string "PrimeSim_XA")
```

- If the default is PrimeSim_Pro, use the following command:

```
envSetVal("asimenv.startup" "simulator" 'string  
"PrimeSim_Pro")
```

- If the default is PrimeSim_HSPICE, use the following command:

```
envSetVal("asimenv.startup" "simulator" 'string "HSPICE")
```

- b. Set the following optional settings:

```
envSetVal("asimenv" "saveDir" 'string "./artist_states")  
envSetVal("asimenv.startup" "projectDir" 'string "./simulation")
```

- c. For Virtuoso version IC23.1 and above (also known as Virtuoso Studio), you can use Large-Scale Cloud Simulation (LSCS) job control mode with PrimeSim and HSPICE.

- If PrimeSim or HSPICE version is older than V-2023.12-SP1, set environment parameter in `.cdsinit`:

```
envSetVal("maestro.lscs" "useLSCSFor3rdPartySimulators" 'boolean  
t)
```

- If PrimeSim or HSPICE version is V-2023.12-SP1 and above, no environment parameter is needed.

3. Launch the Virtuoso. The following temporary links are created in the working directory.

```
PrimeSim_SPICE.cdsenv  
PrimeSim_Pro.cdsenv  
PrimeSim_XA.cdsenv  
HSPICE.cdsenv
```

These are the default settings of PrimeSim environment variables.

- To generate the temporary links, set `CDS_LOAD_ENV` to `CSF`. These temporary links are automatically deleted when you exit **Virtuoso**.
- To get a complete list of options and environment variables, use the following commands in the **Virtuoso** console:

Note:

Take `PrimeSim_SPICE` as example.

- To get a list of environment variables (`PrimeSim_SPICE.envOpts`):

```
asiDisplayEnvOption(asiGetTool('PrimeSim_SPICE))
```

- To get a list of simulator options (`PrimeSim_SPICE.opts`):

```
asiDisplaySimOption(asiGetTool('PrimeSim_SPICE))
```

- To get a list of output options (PrimeSim_SPICE.outputs):

```
asiDisplayKeepOption(asiGetTool('PrimeSim_SPICE'))
```

- To get a list of options in tran analysis (PrimeSim_SPICE.<analysisName>):

```
asiDisplayAnalysisField(asiGetAnalysis(asiGetTool('PrimeSim_SPICE') 'tran'))
```

```
asiDisplayAnalysisOption(asiGetAnalysis(asiGetTool('PrimeSim_SPICE') 'tran'))
```

- To override the default of environment variables, set the parameters in the local .cdsinit or .cdsenv file. The priority is in the following order:

```
.cdsinit > .cdsenv > PrimeSim_SPICE.cdsenv / PrimeSim_XA.cdsenv /  
PrimeSim_Pro.cdsenv / HSPICE.cdsenv
```

If you set the variables in the .cdsinit file, set it after the line of load (".../snpsSimADE.ile").

For example: To set up the model file in the .cdsinit file,

```
envSetVal("PrimeSim_SPICE.envOpts " "modelFiles" 'string  
"/path/model.scs")
```

After the .cdsinit is set, you can launch Virtuoso.

4. Verify the PrimeSim ADE Integration interface installation.

If the PrimeSim ADE Integration interface is installed correctly, the following messages are displayed in the Virtuoso console:

```
==== snpsSimADE Interface, BUILD Number (Build Date, Build Time) ====  
==== COPYRIGHT 2008-2024 SYNOPSYS INC.  
====  
snpsSimADE: Loading context at  
'("/opt/... /interface/spiceADE/snpsSimADE.ile")  
snpsSimADE: Register HSPICE ...  
snpsSimADE: use HSPICE at  
'/global/apps/hspice_2023.12-2/hspice/bin/hspice'  
snpsSimADE: HSPICE Version V-2023.12-2 linux64 (Jan 31 2024 8697540)  
snpsSimADE: Register FineSim ...  
snpsSimADE: use FineSim at  
'/global/apps/finesim_2023.12-2/bin/finesim'  
snpsSimADE: FineSim V-2023.12-2 for linux64 (Feb 04, 2024)  
snpsSimADE: Register PrimeSim_XA ...  
snpsSimADE: use xa at '/global/apps/xa_2023.12-2/bin/xa'  
snpsSimADE: PrimeSim (TM) XA Version V-2023.12-2 for linux64 - Feb  
01, 2024  
snpsSimADE: Register PrimeSim_SPICE ...  
snpsSimADE: use PrimeSim SPICE at '/opt/... //bin/primesim_spice'
```

Chapter 2: Installation

Installing the PrimeSim ADE Integration Interface

```
snpsSimADE: PrimeSim SPICE V-2023.12 for linux64 (Feb 26, 2024)
Loading spectrei.cxt
snpsSimADE: Register PrimeSim_Pro ...
snpsSimADE: use PrimeSim Pro at '/opt/... //bin/primesim_pro'
snpsSimADE: PrimeSim Pro V-2023.12 for linux64 (Feb 26, 2024)
==== snpsSimADE Interface installed successfully
=====
```

3

Basic Simulation Flow for PrimeSim SPICE

To set up a basic simulation flow for PrimeSim SPICE, perform the following steps:

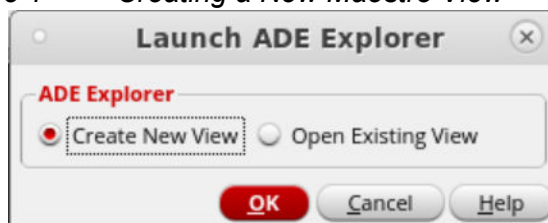
1. Create a new maestro view:

Note:

PrimeSim_SPICE is the default simulator.

- a. Choose **Launch > Launch ADE Explorer > Create New View > OK**.

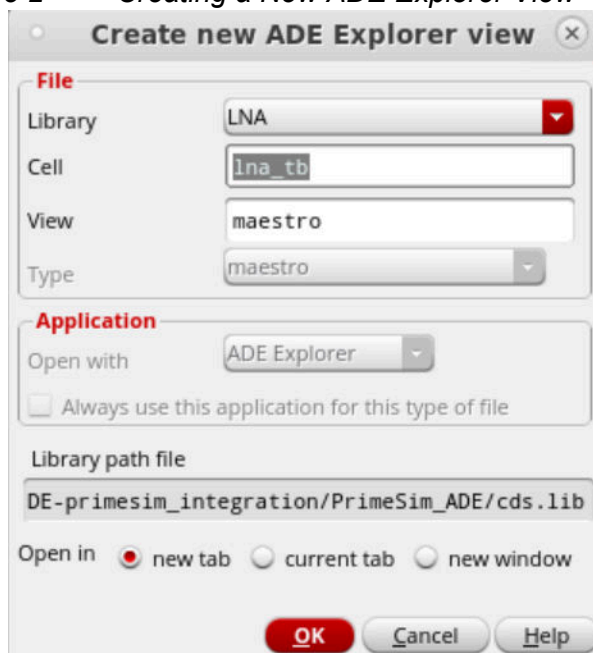
Figure 1 Creating a New Maestro View



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- b. Select the value for **Library** and type the **Cell** value. Click **OK**.

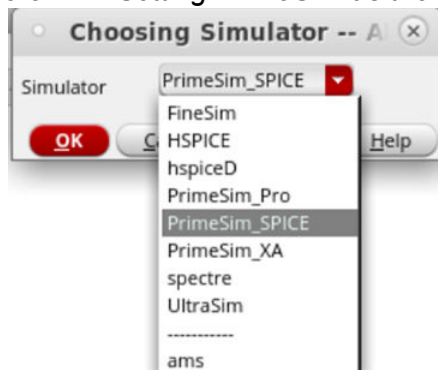
Figure 2 Creating a New ADE Explorer View



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2. To open an existing maestro view that is created using the third-party simulator:
 - a. Choose **Setup > Simulator** to open the **Choosing Simulator** dialog box.
 - b. For **Simulator**, select **PrimeSim_SPICE**, **PrimeSim_Pro**, **PrimeSim_XA**, or **HSPICE**.

Figure 3 Setting PrimeSim as the Default Simulator



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Note:

- On switching the simulator in maestro view, ADE-integration flow maps most of the options automatically.
- If the options do not map, you can check the mapping report in console. It gives a list of unmapped options.

Figure 4 Checking the Option Mapping Report

```
=====
HARMONIC BALANCE
=====
```

NAME	PROMPT	VALUE	REASON
sweepB1	sweepB1	t	No matching option
sweepBcheck3	sweepBcheck3	t	No matching option
sweepBcheck1	sweepBcheck1	t	No matching option
sweepBcheck2	sweepBcheck2	t	No matching option

```
=====
```

3. Set up the job policy. To do this, perform the following steps:

- From the ADE menu, select **Setup > Job Policy Setup**.

This opens the ADE-native **Job Policy Setup** form.

Figure 5 Setting the Job Policy

The screenshot shows the 'Job Policy Setup' dialog box for 'LNA_Ina_tb_dc_1 (on us01odc-sc5)'. The 'Simulation' tab is selected. At the top, there's a 'Job Policy Name' dropdown menu set to 'Maestro Default', with 'Save...' and 'Delete' buttons. Below this, the 'Setup' section contains 'Job Control Mode' with radio buttons for 'LSCS' (selected) and 'ICRP', 'Distribution Method' with a dropdown set to 'Local', and 'Max. Jobs' with a text box containing '1'. The 'Timeouts (in Secs.)' section has 'Start Timeout' (300), 'Simulation Run Timeout' (empty), and 'Linger Time' (300). The 'Error reporting' section has 'Show output log on error' (unchecked) and 'Show errors even if retrying test' (checked). At the bottom are 'OK', 'Cancel', 'Apply', 'Stop All', and 'Help' buttons.

- b. In the **Simulation** tab, select the **Job Policy Name** from the drop-down menu.

4. Set up the simulation options. To do this, select **Setup > High Performance Simulation** from the ADE menu.

This opens the **High-Performance Simulations Options** form, where you can set the required options.

Figure 6 *Setting High-Performance Simulation Options*

High-Performance Simulation Options (on us01odc-sc5-1-pbm)

Accuracy -

Accuracy Level ☐ Default ☒ runlvl

☐ 1 ☐ 2 ☐ 3 ☒ 4 ☐ 5 ☐ 6

Tolscale

Max Time Step (delmax)

Convergence -

Convergence Level ☒ 0 ☐ 0.5 ☐ 1 ☐ 2 ☐ 3

Post Layout Simulation -

Rcred ☐ 0 ☐ 1 ☐ 2 ☐ 3 ☒ Default

Number of Threads (-np) Simulator Options

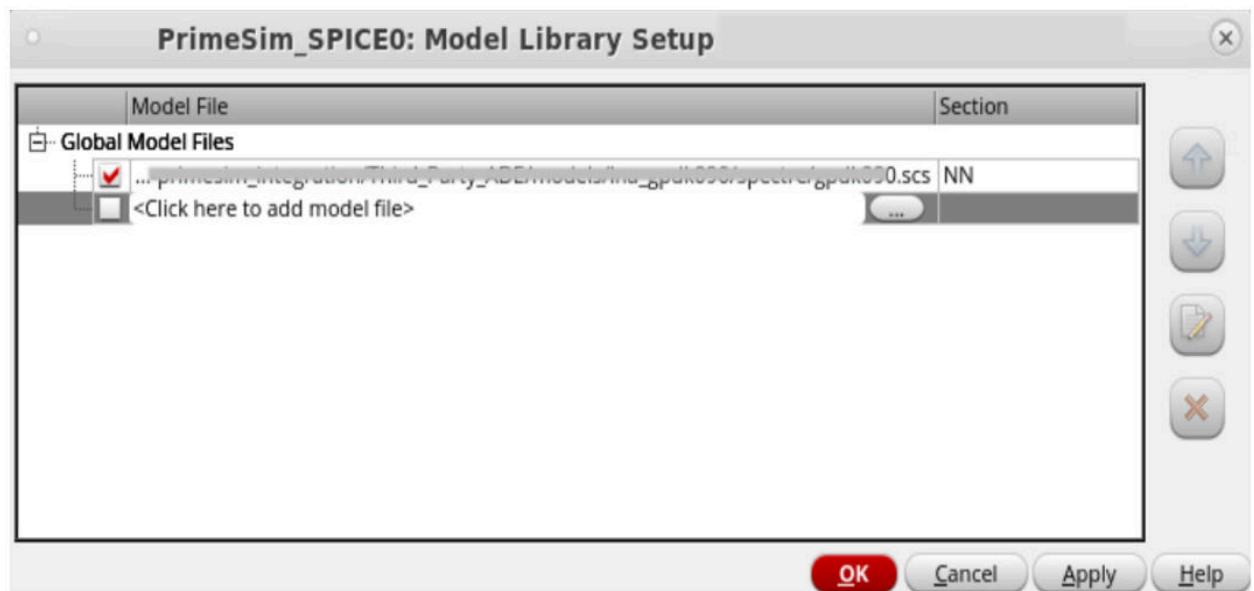
Number of GPUs (-ngpu)

Use LSF_HPC ☐

OK Cancel Apply Help

5. Set up the model library. To do this, select **Setup > Model Library Setup** from the ADE menu.

Figure 7 Setting the Model Library



6. Use the FSDB format in Virtuoso and PrimeSim.

- When using PrimeSim inside ADE, you are recommended to use `psf` format.

Note:

You can set FSDB as the output format, for transient results with large data size.

- Virtuoso has a build-in command to show the latest FSDB version it can support. In the Virtuoso terminal, type the command below:

```
srrversion -fsdb
```

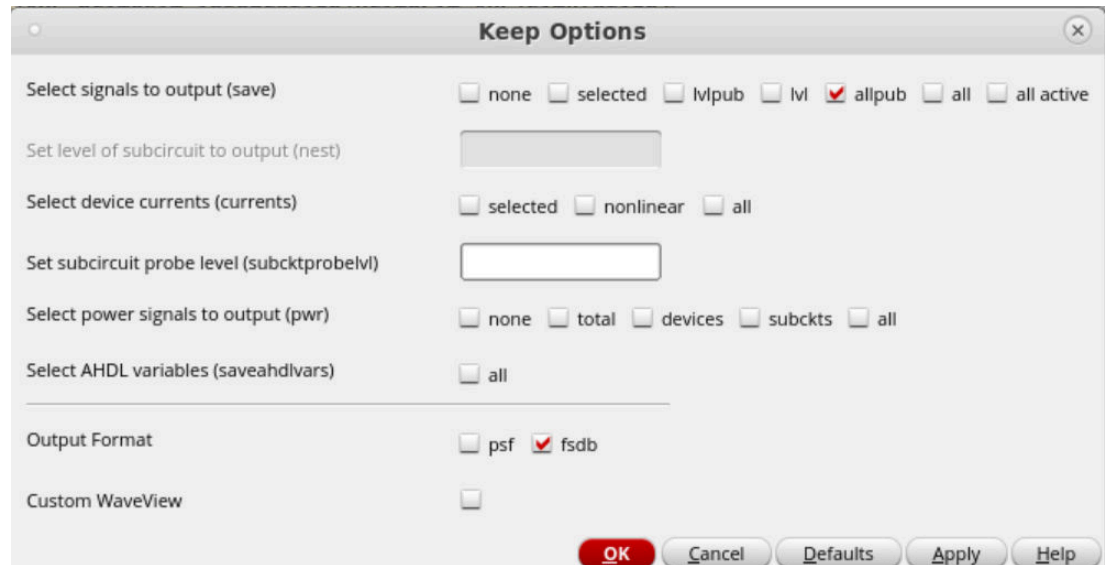
An example result from IC 23.10 (Virtuoso Studio) is:

```
fsdb-rapi : 6.3
fsdb-wapi : 6.3
fsdb-kits: FsdBReader/23.03-p001/lxx86 FsdBWriter/*/lxx86
FsdBIndep/*/indep:lxx86 FsdBReader/23.03-p001/lxa64
FsdBWriter/*/lxa64
```

- To setup the FSDB format in ADE, do the following steps:

- a. In the ADE menu, select **Output > Save Options**.

The **Keep Options** window opens.

Figure 8 *Setting the FSDB Format in ADE*

b. You can set to use **WaveView** on this window too.

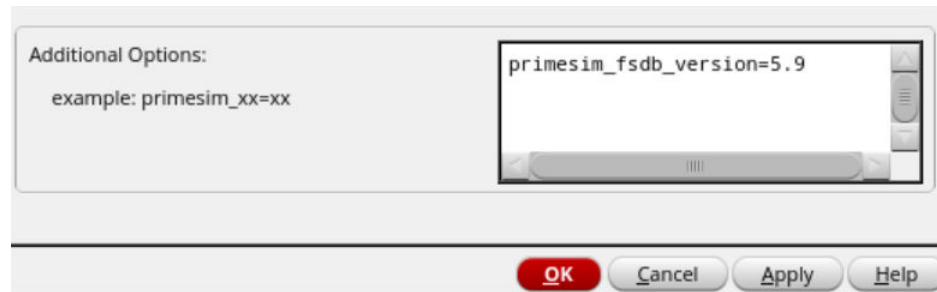
Note:

- Only use FSDB format if the transient result size is too big, as the plotting is too slow in PSF format.
 - For most of the cases, just use the default PSF format.
- If using PrimeSim V-2023.12-SP2 with Virtuoso Studio, the FSDB version is supported by default. No need to set the version number for FSDB format.

- To set the FSDB version, do the following steps:
 - a. From the ADE menu, select **Simulation > Analog**.
 - b. Add the following option in **Additional Options**:

```
primesim_fsdb_version=<$version>
```

Figure 9 *Adding the FSDB Version*

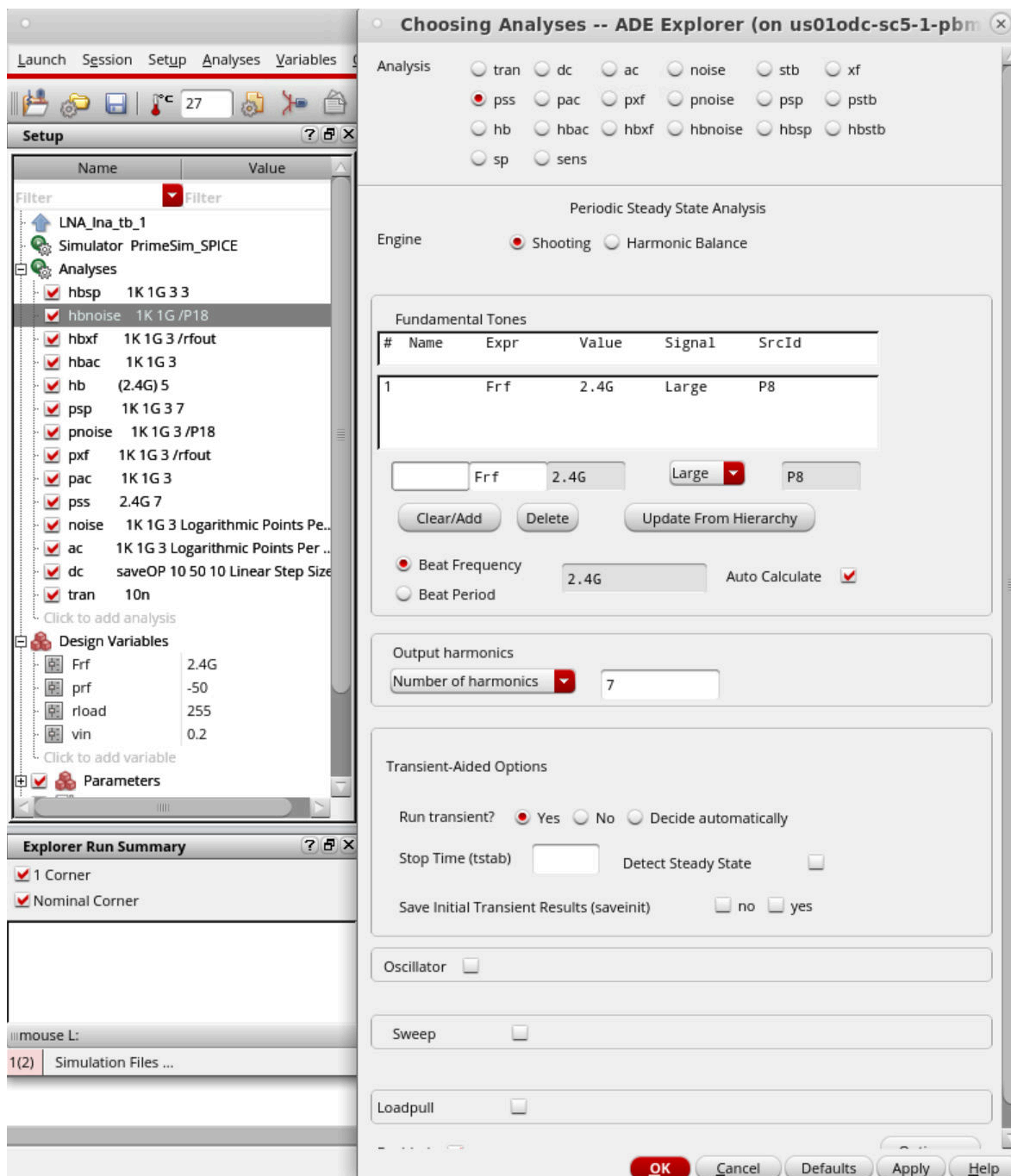


7. Select the analyses. To do this, select **Analyses > Click to add analysis** in the **Setup** view of ADE Explorer.

This pops out the **Choosing Analyses** form.

- This form is designed for PrimeSim and PrimeSim HSPICE simulators.
- This form supports all commonly used analyses types.

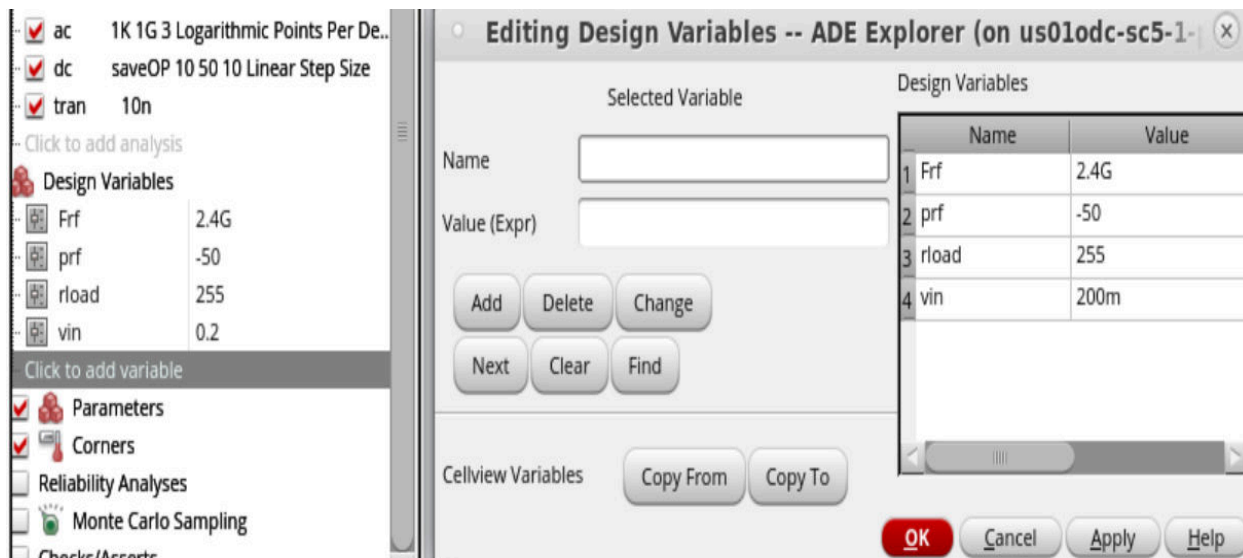
Figure 10 Choosing the Analyses



8. Edit the design variables. To do this, select **Design Variables > Click to add variable** in the **Setup** view of ADE Explorer.

This opens the **Editing Design Variables** form.

Figure 11 Editing the Design Variables

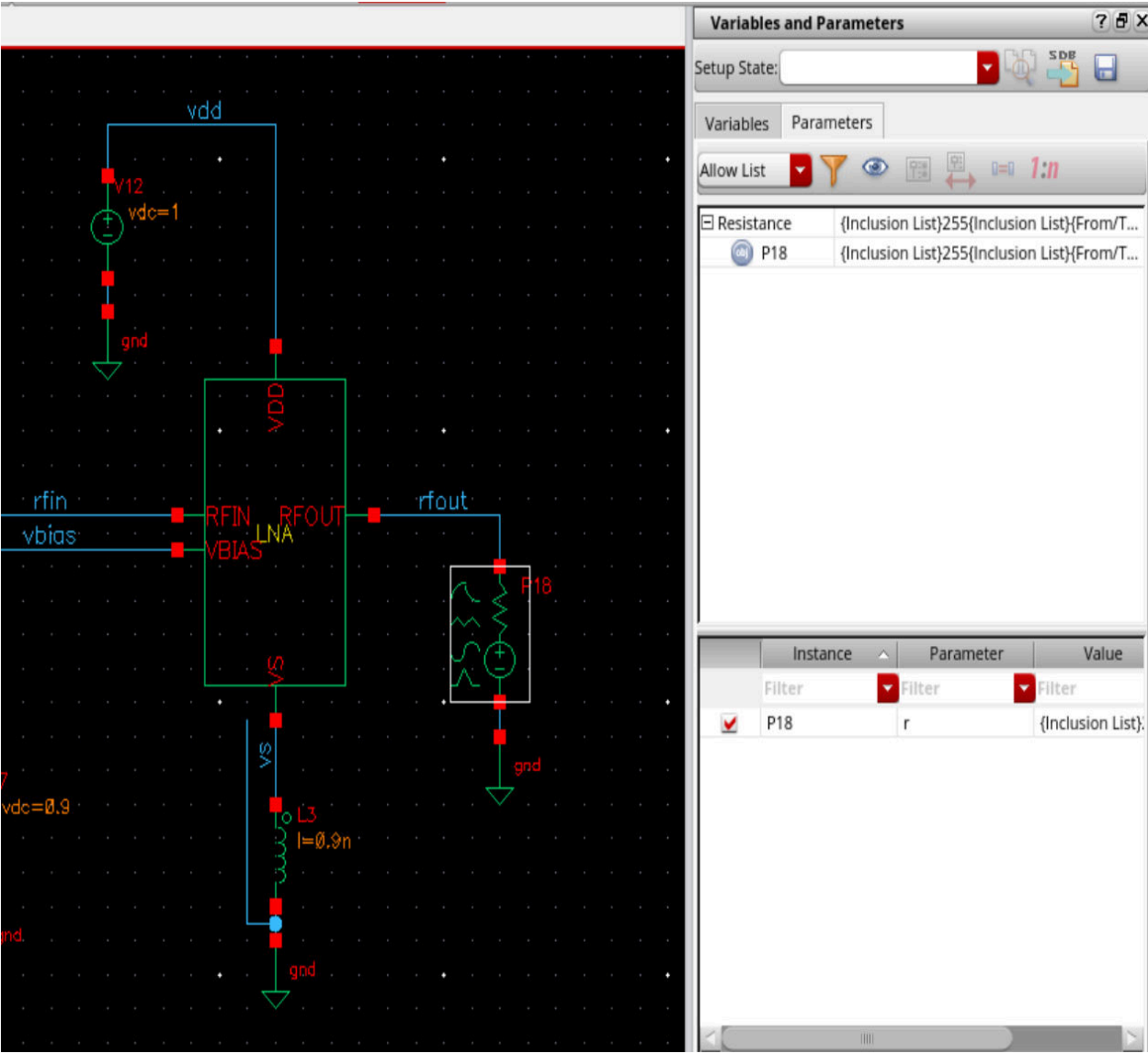


9. Set up the parameters. To do this, perform the following steps:
 - a. Select **Parameter > Click to add parameter** in the **Setup** view of ADE Explorer.

This opens the schematic and pops out the **Variables and Parameters** form.
 - b. Select an instance from the schematic.

This pops out the table of the parameters in the **Variables and Parameters** form.
 - c. Select the instance parameter and create a design parameter from it.

Figure 12 *Setting the Variables and Parameters on the Schematic*



The selected parameters automatically add to the maestro view.

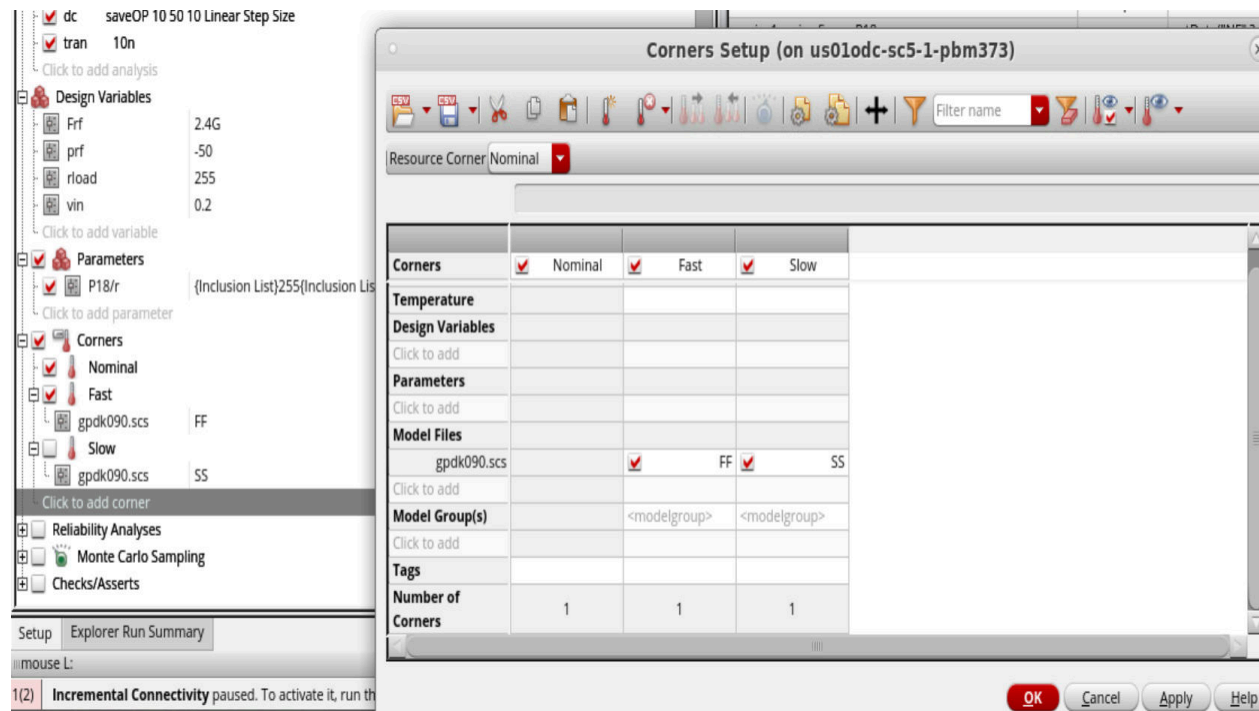
Figure 13 *Parameter Added to the Maestro View*



10. Set up the corners. To do this, select **Corners > Click to add corner** in the **Setup** view of ADE Explorer.

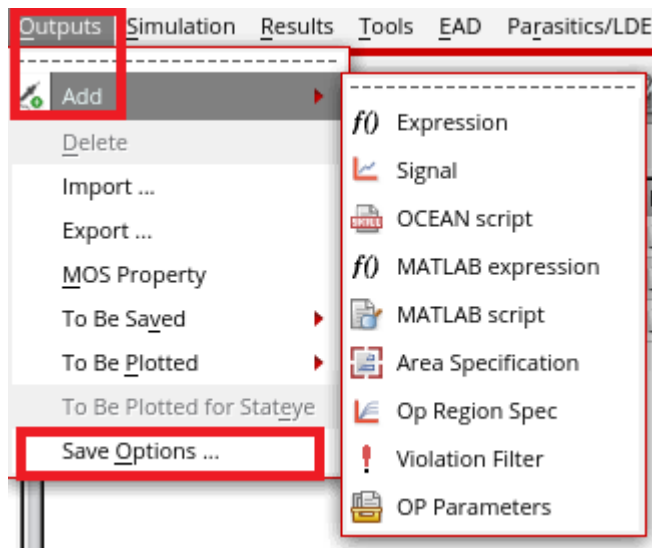
This opens the **Corners Setup** form.

Figure 14 *Setting the Corners*



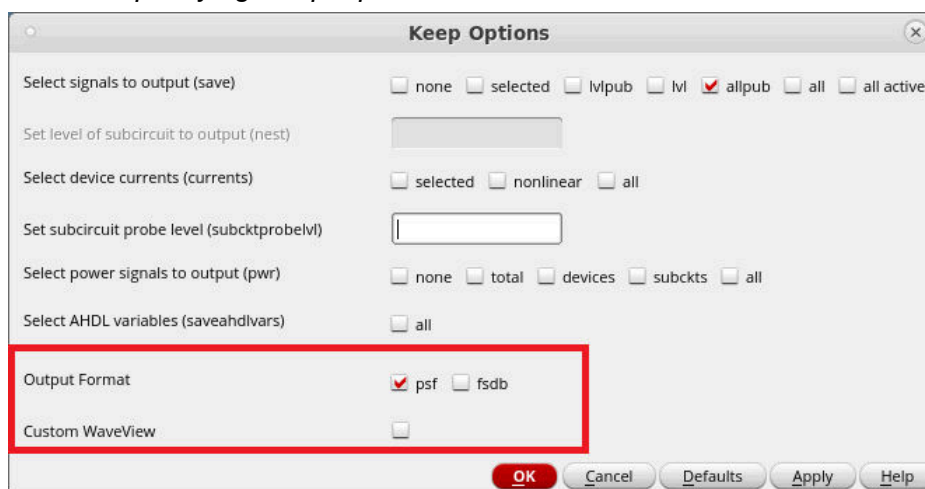
11. Setup the outputs.
- To add any function, output, or signal, select **Outputs > Add**.
 - To save the outputs, select **Outputs > Save Options**.

Figure 15 Adding and Saving Outputs



- Specify the following options in **Keep Options** pane:
 - **Output Format:** Select **psf** or **fsdb**, for PSF and FSDB generation respectively.
 - **WaveView:** Select this option for supporting WaveView.

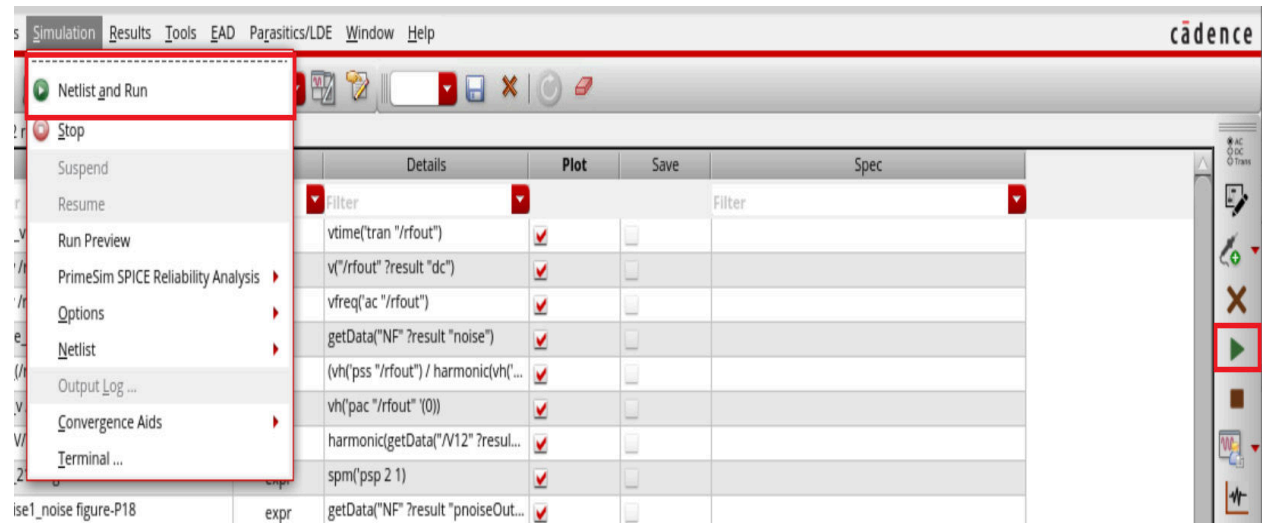
Figure 16 Specifying Keep Options



12. Run the simulation. You can run the simulation in one of the following ways:

- a. From the ADE toolbar, click  **Run Simulation**.
- b. From the ADE main menu, select **Simulation > Netlist and Run**.

Figure 17 Running the Simulation



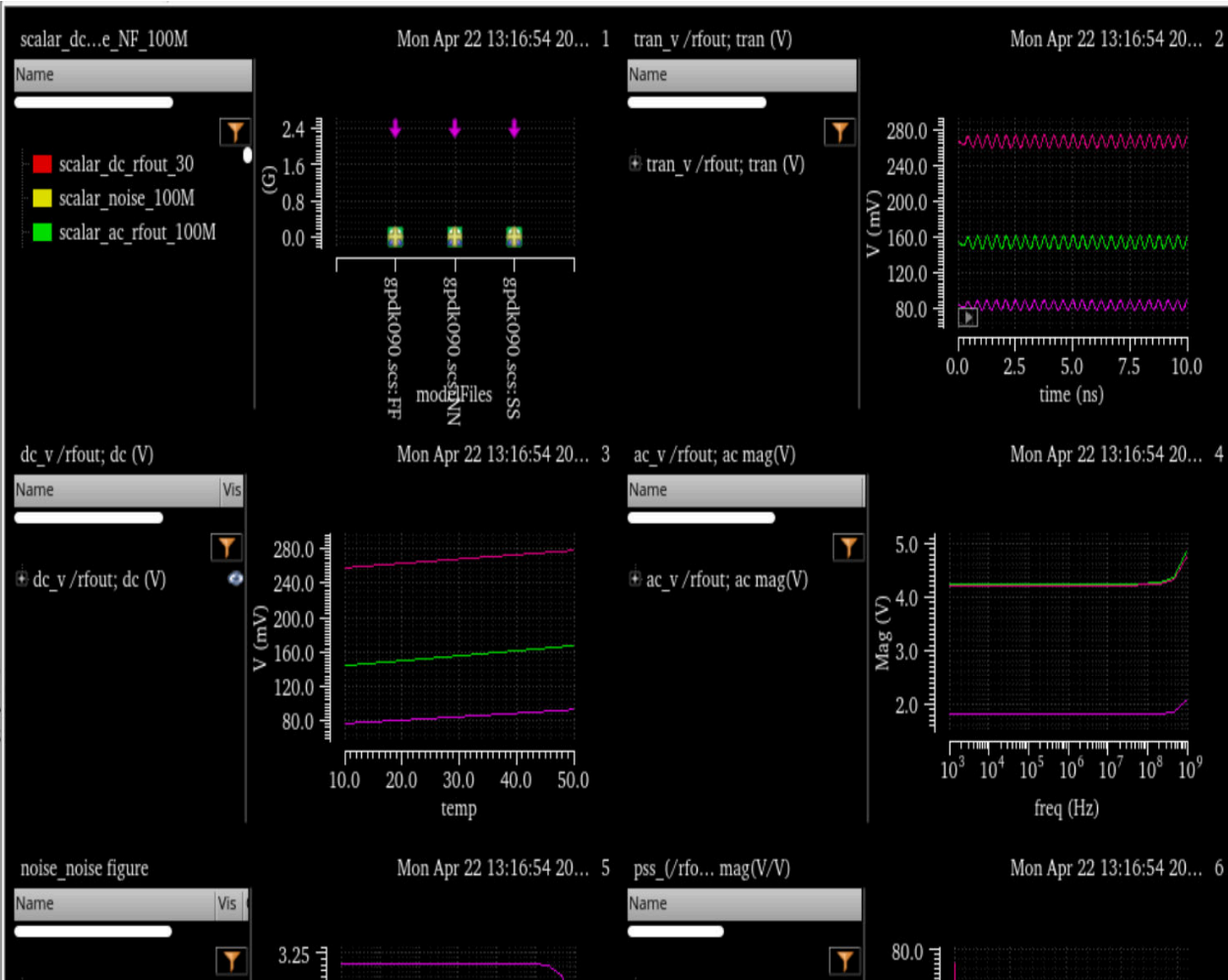
13. Check the simulation results in the output table.

Figure 18 Checking the Results

Point	Test	Output	Nominal	Spec	sig	Pass/Fail	Min	Max	Fast	Slow
Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter
1	LNA_ina_tb_1	hbsp_S11 dB20								
1	LNA_ina_tb_1	hbxfr_V /V12; hb...								
1	LNA_ina_tb_1	hbnoise1_noise...								
1	LNA_ina_tb_1	scalar_dc_rfou...	155.6m	< 156m		fail	84.7m	267.9m	84.7m	267.9m
1	LNA_ina_tb_1	scalar_noise_10...	2.071	< 2		fail	2.071	3.159	3.159	2.643
1	LNA_ina_tb_1	scalar_ac_rfou...	4.257	> 4		fail	1.816	4.257	1.816	4.23
1	LNA_ina_tb_1	scalar_tran_rfou...	160.5m	< 160m		fail	88.83m	272.8m	88.83m	272.8m
1	LNA_ina_tb_1	scalar_tran_rfou...	160.5m	< 160m		fail	88.83m	272.8m	88.83m	272.8m
1	LNA_ina_tb_1	scalar_tran_freq	2.4G	maximize 2.5G		near	2.4G	2.4G	2.4G	2.4G
1	LNA_ina_tb_1	scalar_tran_std...	165.6u	< 3.5u		fail	165.6u	165.6u	eval err	eval err
1	LNA_ina_tb_1	scalar_pss_vgain...	2.176	minimize 2.3		pass	1.024	2.176	1.024	2.059
1	LNA_ina_tb_1	scalar_pac_rfou...	367.4m	maximize 370m		fail	171.1m	561.6m	171.1m	561.6m
1	LNA_ina_tb_1	scalar_pxf_V12...	8.139m	< 10m		fail	1.41m	11.33m	11.33m	1.41m
1	LNA_ina_tb_1	scalar_psp_S21...	2.482	> 2		near	1.87	2.482	1.87	2.473
1	LNA_ina_tb_1	scalar_pnoise1...	29.41	< 30		fail	18.19	42.95	18.19	42.95
1	LNA_ina_tb_1	scalar_hb_v /rfo...	-42.01	> -40		fail	-45.95	-42.01	-45.95	-42.59
1	LNA_ina_tb_1	scalar_hbac_gai...	43.69	< 44		fail	35.58	54.46	35.58	54.46
1	LNA_ina_tb_1	scalar_hbsp_S11...	-2.26	> -2.3		fail	-2.709	-2.26	-2.709	-2.334
1	LNA_ina_tb_1	scalar_hbxfr_100...	-12.02				-18.83	-9.178	-18.83	-9.178
1	LNA_ina_tb_1	scalar_hbnoise...	29.4				18.17	43.01	18.17	43.01
Parameters: P18.r=245										
2	LNA_ina_tb_1	tran_v /rfout; tra...								
2	LNA_ina_tb_1	dc_v /rfout; dc (V)								
2	LNA_ina_tb_1	ac_v /rfout; ac m...								
2	LNA_ina_tb_1	noise_noise figure								
2	LNA_ina_tb_1	pss_v /rfout) /rfi...								
2	LNA_ina_tb_1	pac_v /rfout h=1...								
2	LNA_ina_tb_1	pac_v /rfout h=1...								

The waveforms are plotted with VIVA or WaveView.

Figure 19 *Plotted Waveforms*



4

Setting up Multiple Technology Simulation

To set up multiple technology simulator (MTS), perform the following steps:

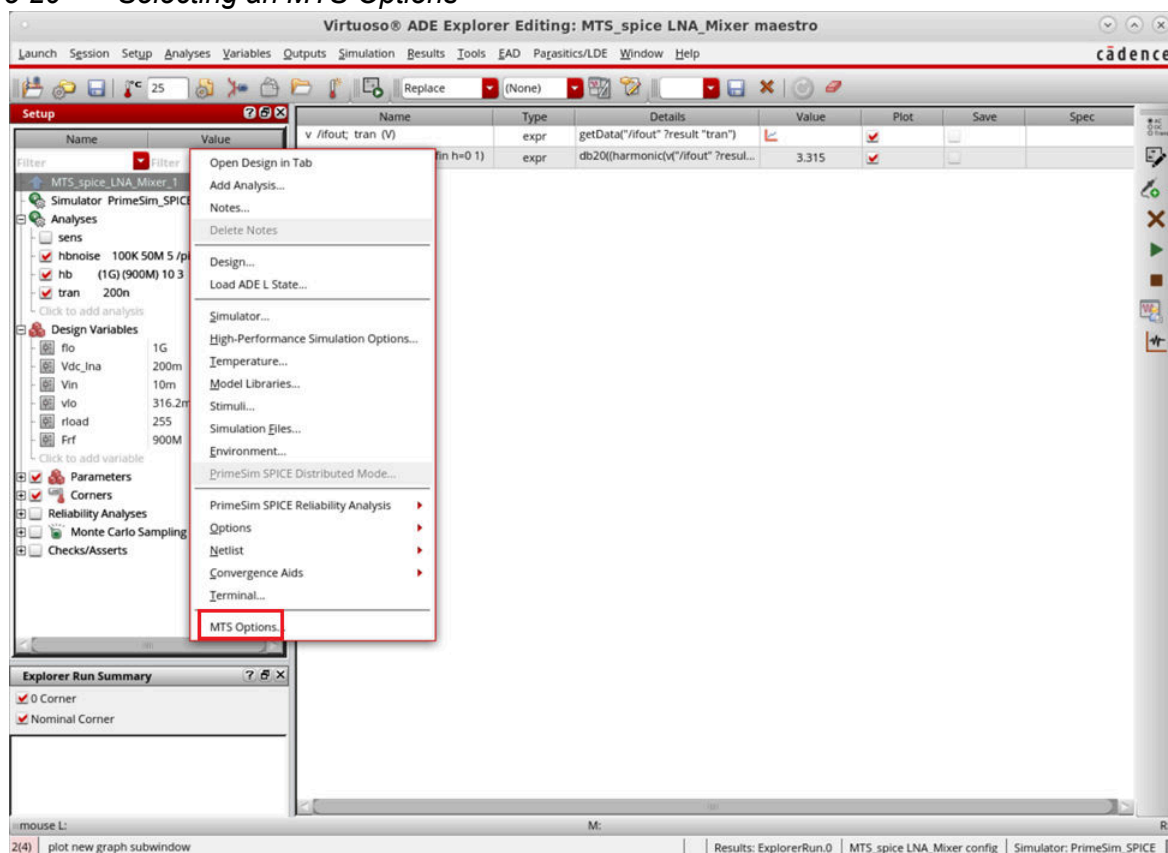
1. Open a maestro view and set **Simulator** to **PrimeSim_SPICE** or **PrimeSim_XA**.

Note:

The design should be configuration view. Schematic view is not supported in MTS flow.

2. Right-click the test name, and select **MTS Options**.

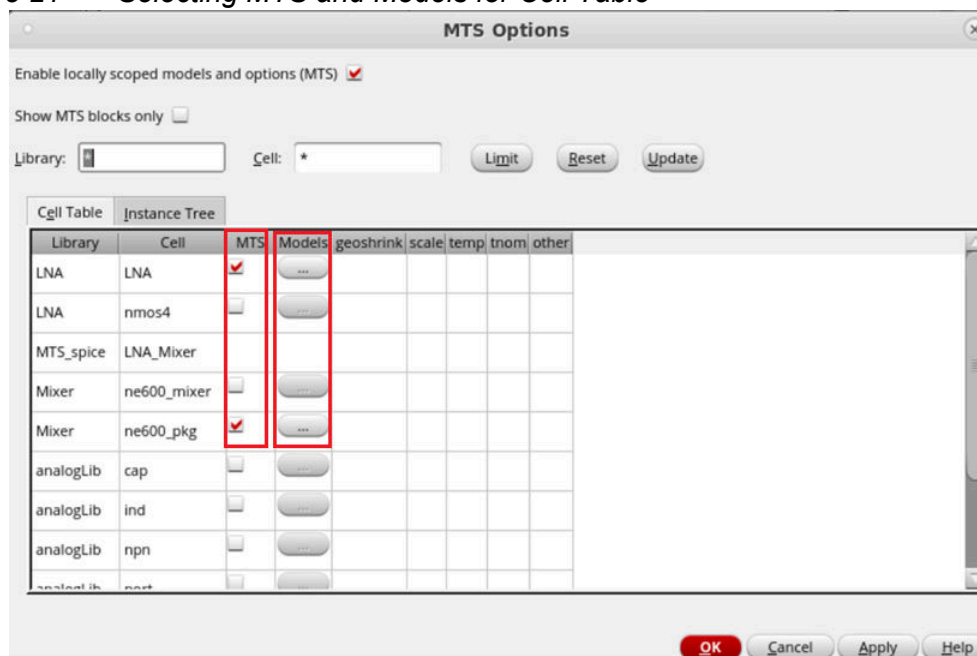
Figure 20 Selecting an MTS Options



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3. Set the MTS using the following methods in **MTS Option** dialog box:
 - a. Cell Table: You can specify the special MTS setting of model files and parameters.
 - i. Click **Cell Table** tab and select the check box in **MTS** column for **LNA** and **Mixer** library.
 - ii. Click the ... button in the **Models** column for **LNA** or **Mixer** Library to open the **Models Library Setup** dialog box.

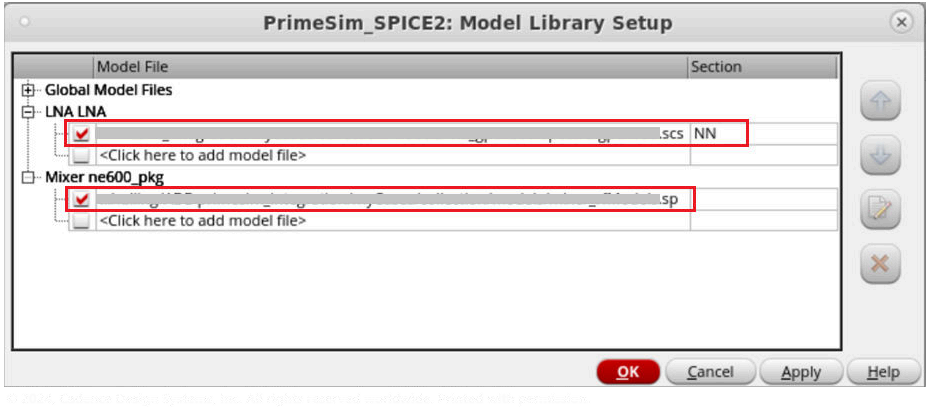
Figure 21 Selecting MTS and Models for Cell Table



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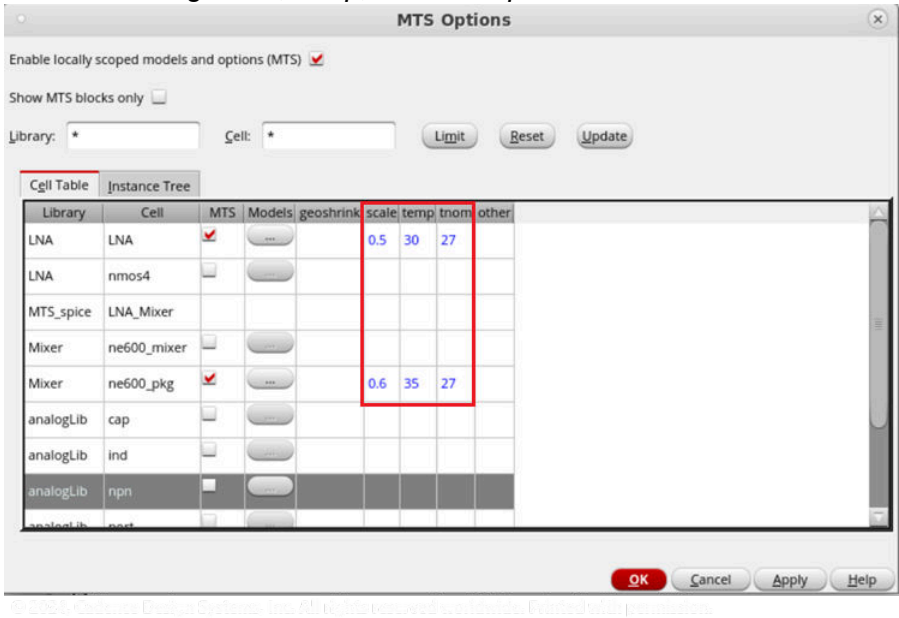
- iii. Select the model files and click **OK**.

Figure 22 *Selecting Model Files for Cell Table*



iv. Add appropriate values for **scale**, **temp**, and **tnom** fields.

Figure 23 *Adding scale, temp, and tnom parameters for Cell Table*



v. After the setup, click **OK** to exit the **MTS Options** dialog box.

vi. Generate netlist and run simulation by standard ADE flow.

Example 1 MTS Netlist for Cell Table

```
// Begin MTS Block
// Library name: LNA
// Cell name: LNA
```

```

// View name: schematic
// Inherited view list: schematic spectre
subckt LNA RFIN RFOUT VBIAS VDD VS
include "mts-LNA-LNA.scs"
    M4 (RFOUT VBIAS net13 net13) gpdk090_nmos1v w=10u l=100.0n
    m=10
    ...
ends LNA
// End of subcircuit definition.
// End of MTS Block

// Begin MTS Block
// Library name: Mixer
// Cell name: ne600_pkg
// View name: schematic
// Inherited view list: schematic spectre
subckt ne600_pkg IF LO RF VDC VDC_LO VDC_RF inh_bulk_n
include "mts-Mixer-ne600_pkg.scs"
...

```

The MTS settings are created as two include files, mts-LNA-LNA.scs and mts-Mixer-ne600_pkg.scs. Open the mts-LNA-LNA.scs file and check the model file and parameter settings:

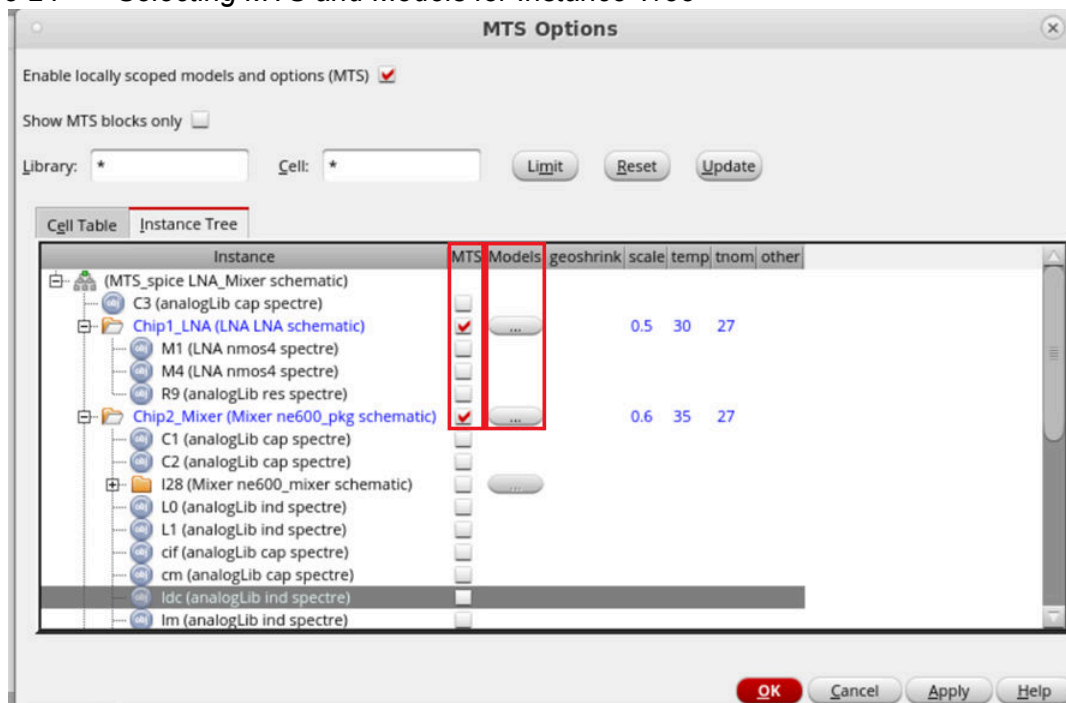
```

include "/remote/... /gpdk090.scs" section=NN
simulatorOptions options scale=0.5 temp=30 tnom=27

```

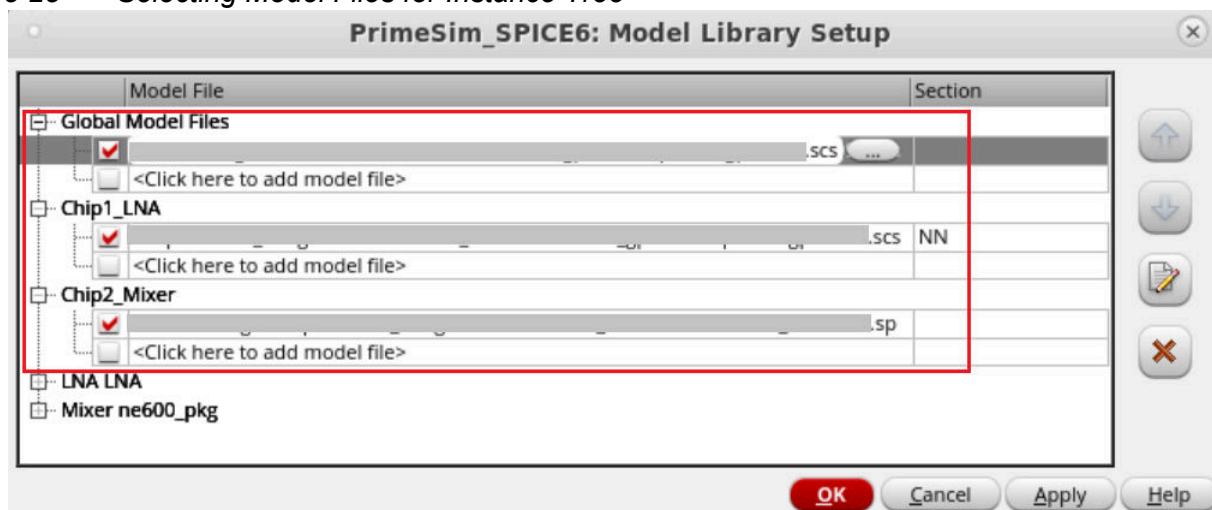
- b. Instance Tree: You can specify instances with more customized settings.
 - i. Click **Instance Tree** tab and select the check box in **MTS** column for **LNA** and **Mixer** library.
 - ii. Click the ... button in the **Models** column for **LNA** or **Mixer** Library to open the **Models Library Setup** dialog box.

Figure 24 Selecting MTS and Models for Instance Tree



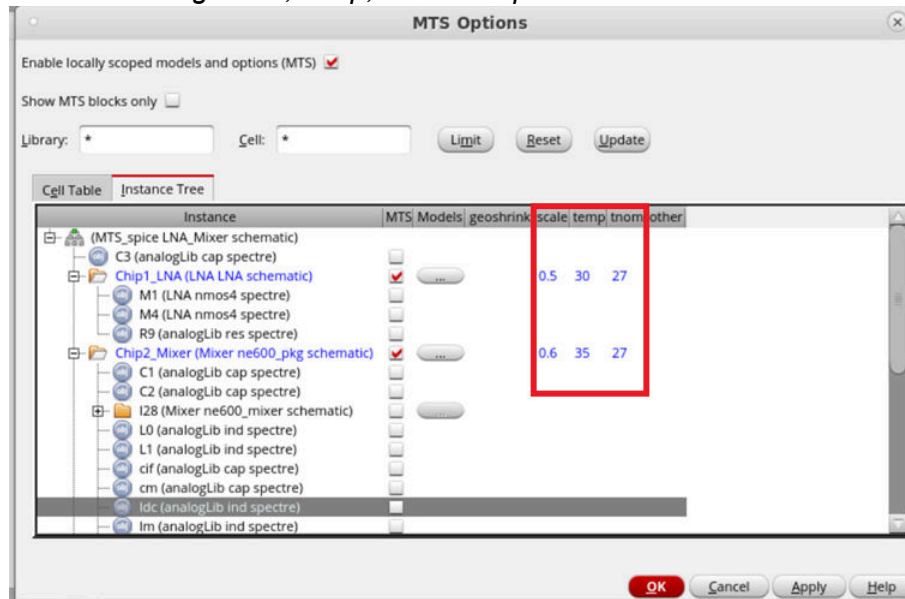
iii. Select the model files and click **OK**.

Figure 25 Selecting Model Files for Instance Tree



iv. Add appropriate values for **scale**, **temp**, and **tnom** fields.

Figure 26 Adding scale, temp, and tnom parameters for Instance Table



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- v. After the setup, click **OK** to exit the **MTS Options** dialog box.
- vi. Generate netlist and run simulation by standard ADE flow.

5

PrimeSim SPICE Monte Carlo

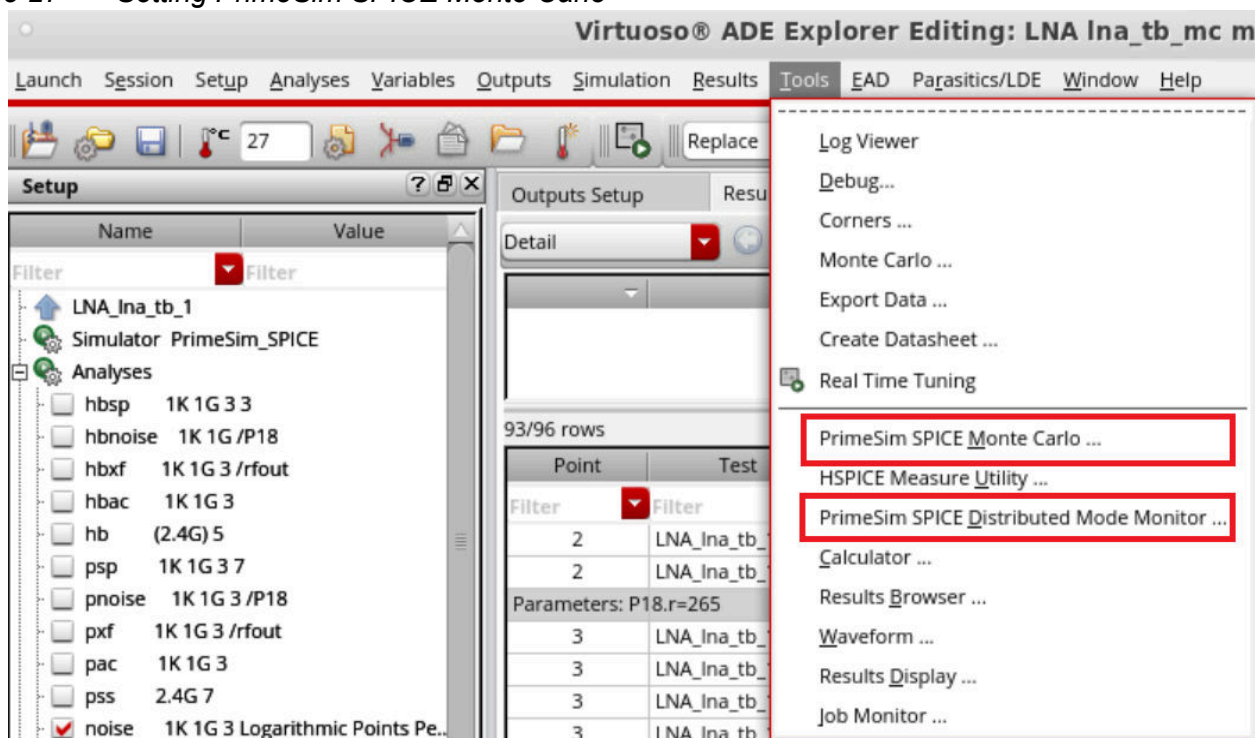
ADE blocks all the third-party simulators from using the native Monte Carlo flow. To set up PrimeSim SPICE Monte Carlo in ADE, perform the following steps:

1. Select **Setup > Model Libraries** from the ADE menu, to select the Monte Carlo model file for the simulation.
2. Select **Tools > PrimeSim SPICE Monte Carlo** from the ADE menu.

Note:

To monitor the distributed jobs, select **Tools > PrimeSim SPICE Distributed Mode Monitor** from the ADE menu.

Figure 27 Setting PrimeSim SPICE Monte Carlo



This opens the **PrimeSim_SPICE Monte Carlo Analysis** form.

3. In the **PrimeSim_SPICE Monte Carlo Analysis** form, do the following steps:
 - a. Set the required Monte Carlo options.
 - b. Select the **Enabled** check-box.
 - c. Click **Apply**.

Figure 28 Setting PrimeSim_SPICE Monte Carlo Analysis

PrimeSim_SPICE Monte Carlo Analysis (on us01odc-sc5-1-pbm373)

File Setup Simulation Results Help

cadence

Setup Variation Outputs Enabled Distribution Max Jobs : 10

Basic

Method ☒ All ☐ Process (Global) ☐ Mismatch (Local)

Number of Points 30 First Point 1

Sampling List ☐

Seed 12345

Sampling Method Sobol

Sampling File

Sigma Scaling Factor (sigma amplification) ☐

Options

Save Waveforms ☒

Run Points for Each Job auto

Variation on Selected Blocks ☐

Enabled ☒

This applies the Monte Carlo setting. You can close the window now.

Note:

To disable the Monte Carlo setting, open the **PrimeSim_SPICE Monte Carlo Analysis** form again, deselect **Enabled**, and click **Apply**.

4. Select **Setup > Job Policy Setup** from the ADE menu.

This opens the **Job Policy Setup** form.

Figure 29 Setting the Job Policy

Job Policy Setup - LNA_Ina_tb_1 (on us01odc-sc5-1-)

Simulation | Netlisting | Resources

Job Policy Name: Save... Delete

Setup

Job Control Mode: ☒ LSCS ☐ ICRP

Distribution Method:

Command:

Max. Jobs:

Timeouts (in Secs.)

Start Timeout:

Simulation Run Timeout:

Linger Time:

Error reporting

Show output log on error: ☐

Show errors even if retrying test: ☒

OK Cancel Apply Stop All Help

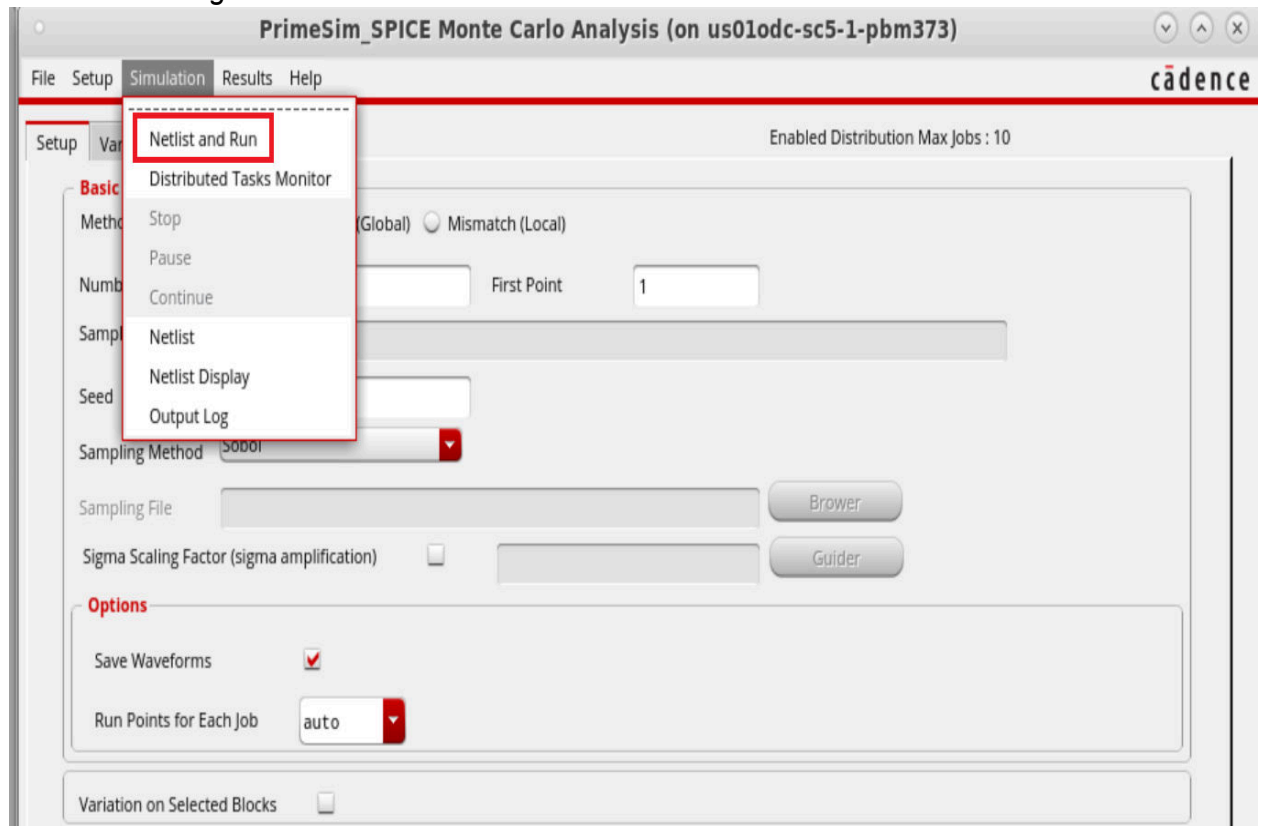
5. In the **Job Policy Setup** form, do the following:
 - a. Select the job policy name from the drop-down menu.
 - b. In the **Command** field, enter the command to run.
 - c. In the **Max. Jobs** field, enter the maximum number of jobs to be run.

Note:

The settings honored by PrimeSim SPICE Monte Carlo are **Command** and **Max. Jobs**. All other settings are ignored in ADE **Job Policy Setup** form.

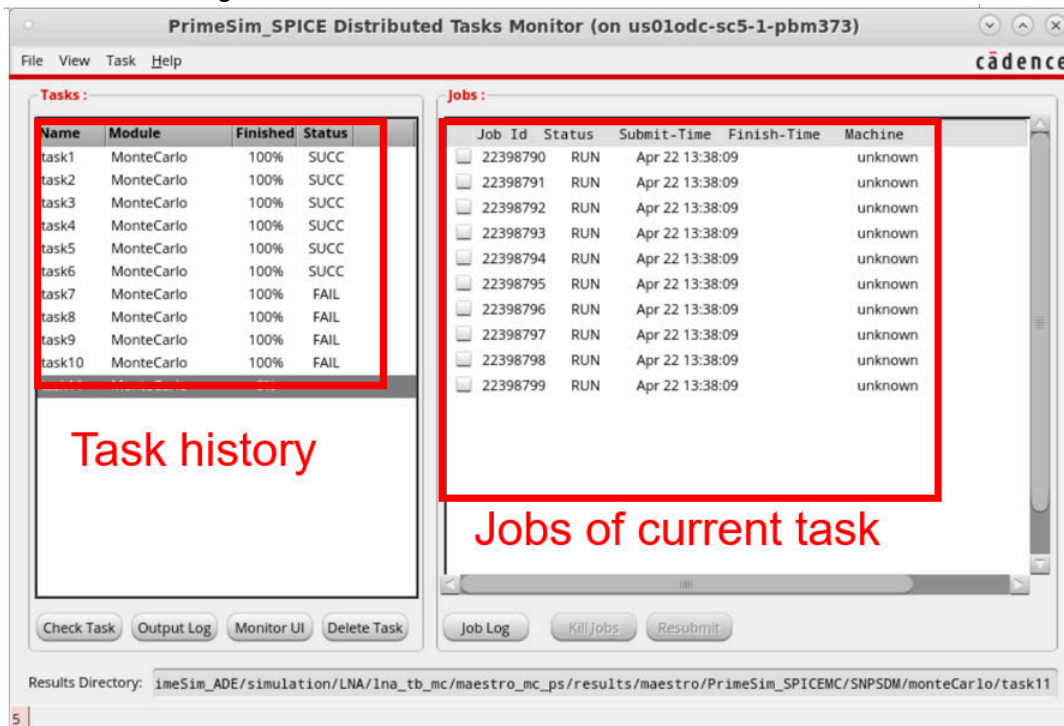
6. Run the Monte Carlo simulation. You can run the simulation in one of the three ways:
 - From the **PrimeSim_SPICE Monte Carlo Analysis** form.
 - From the ADE-native **Run Simulation** button.
 - From the ADE menu, **Simulation > Netlist and Run**.

Figure 30 Running the Monte Carlo Simulation



7. Monitor the distributed jobs. You can check the status of distributed jobs on **PrimeSim_SPICE Distributed Tasks Monitor**, which pops out automatically when running the simulations.

Figure 31 Checking the Status of Distributed Jobs

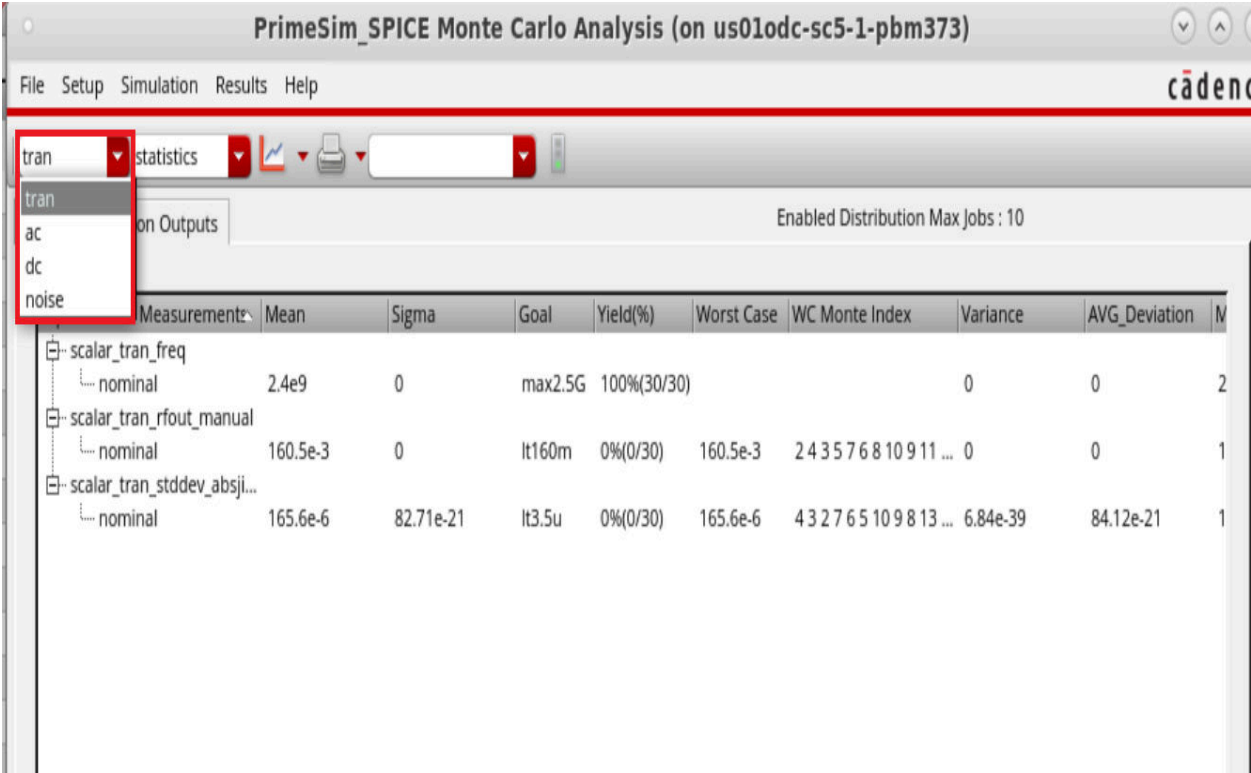


- The form also displays the status of previous tasks.
 - You can refresh the form to check the status updates.
8. Check the variation outputs.

After the simulation, you can see the variation outputs displayed in the table and in the histogram waveforms.

- In the table, you can switch the analyses types to check the variation of different types of analyses.

Figure 32 Checking the Variation Outputs



- The histogram plots are popped out automatically in VIVA. To change the plot type in histogram, select the **Plot** drop-down menu in **PrimeSim_SPICE Monte Carlo Analyses** form.

Figure 33 *Selecting the Chart Plot Type for Histogram*

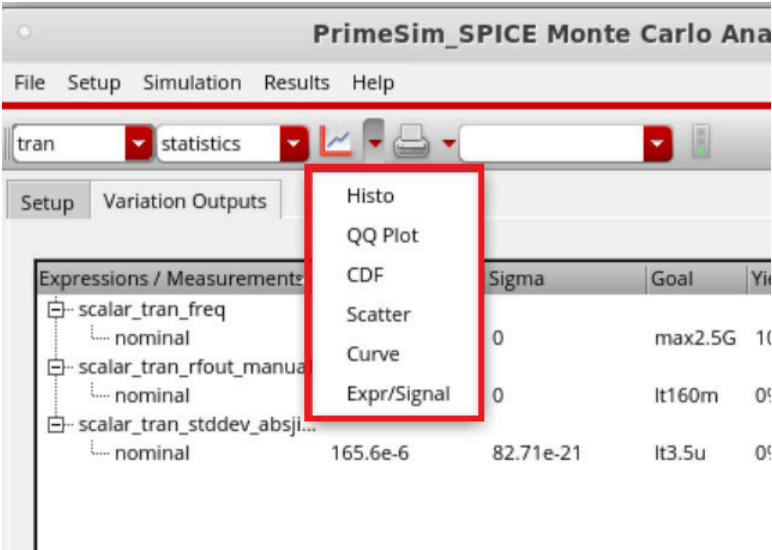
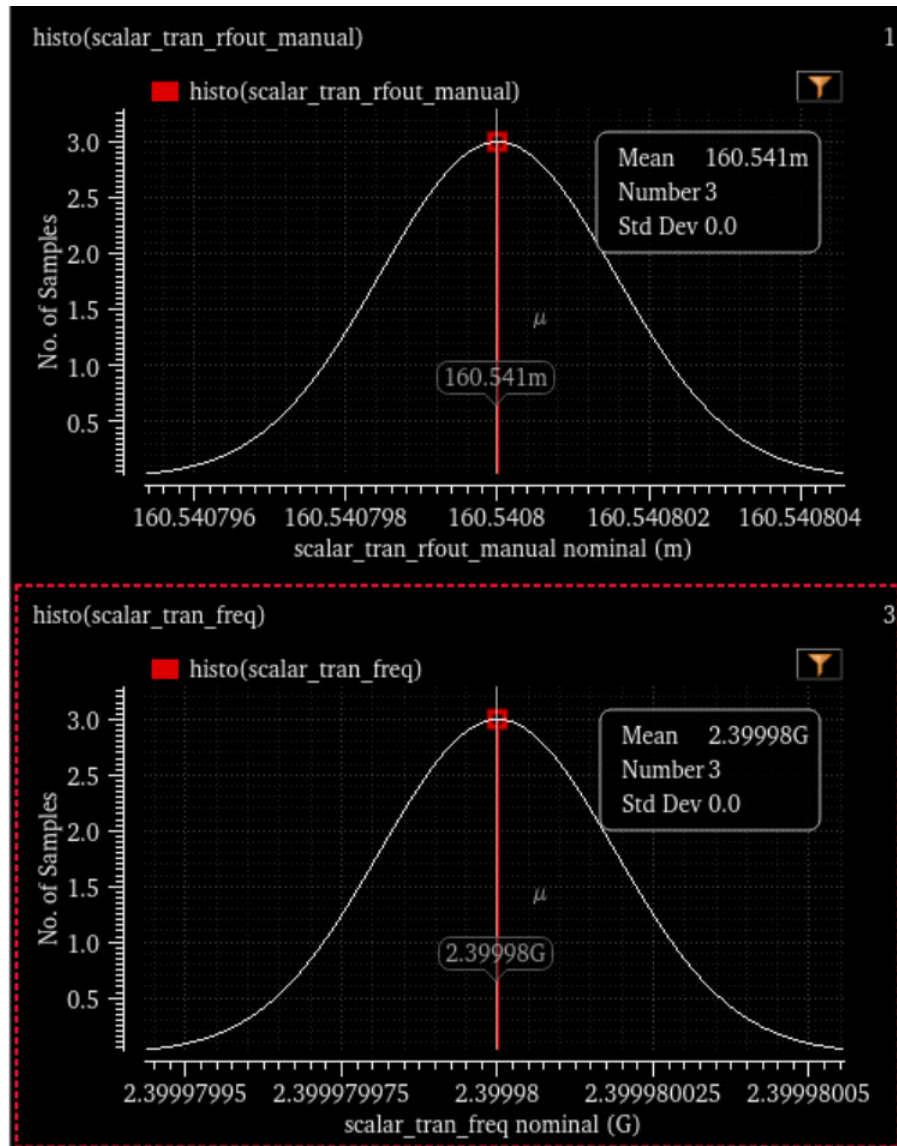


Figure 34 Checking the Histogram



Limitation of PrimeSim SPICE Monte Carlo in ADE

The limitations of PrimeSim Monte Carlo in ADE are:

- PrimeSim SPICE Monte Carlo form does not support ADE Assembler.
- Post processing of Monte Carlo does not support RF analyses.

- Monte Carlo does not support sensitivity analysis for each device's contribution to mismatch.
- You must add the output expressions before the simulation. After the simulation, you cannot evaluate the variation of new added expressions.
- To get the full capability of PrimeSim SPICE Monte Carlo, it is recommended to use Synopsys PrimeWave.

6

PrimeSim SPICE TMI Aging Analyses

ADE blocks all the third-party simulators from using native Reliability Analyses setup.

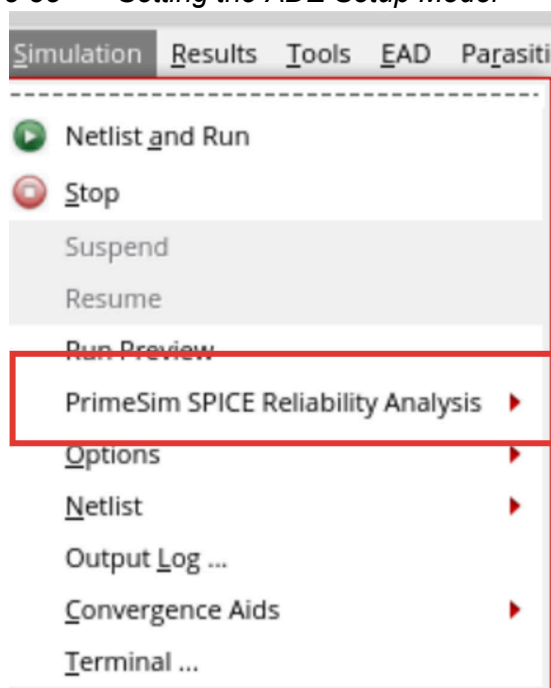
To set up PrimeSim SPICE TMI Aging Analysis in ADE, perform the following steps:

1. From the ADE menu, select **Setup > Model Libraries**.

You can set up the aging model files here.

2. From the ADE menu, select **Simulation > PrimeSim SPICE Reliability Analysis**.

Figure 35 Setting the ADE Setup Model



This opens the PrimeSim SPICE Reliability Form.

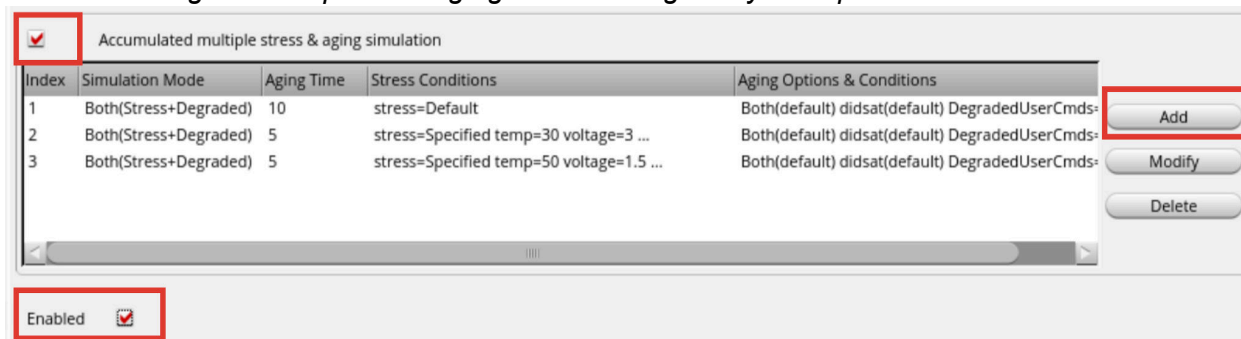
3. Set up single TMI Aging/ Self Heating analyses options.
 - a. Open the **TMI Aging/ Self Heating** window.
 - b. Select the **Setup** tab.

- c. Set up the required options.

Figure 36 *Setting the TMI Aging/ Self-Heating Options*

4. Set up multiple TMI Aging/ Self Heating analyses options.
 - a. Check **Accumulated multiple stress & aging simulation** check-box.
 - b. Click **Add** to add the settings of previous page as new simulation.
 - c. Check **Enabled** check-box.

Figure 37 Setting the Multiple TMI Aging/ Self Heating Analyses Options



- d. Repeat the steps to add new simulation settings.
5. Run the simulation.

- a. Open the TMI Aging/ Self Heating window.
- b. Select the **Enabled** check-box.
- c. Click **Apply** or **OK**.

This applies the TMI Aging / Self Heating analysis setting. You can close the window now.

Note:

To disable the setting, open the TMI Aging/ Self Heating window again, deselect **Enabled**, and click **Apply** or **OK** again.

- d. Use the ADE- native **Run Simulation** button to generate the netlist and run the simulation.

After the simulation, the TMI/Aging settings are written to the end of netlist.

Figure 38 *Running the Simulation*

```
saveOptions options save=allpub
simulator lang=spice insensitive=no

* TMI Aging Simulation .....
* Fresh Simulation & Stress Simulation
.option tmiAge=1
* Degraded Simulation
.alter $AgingTime=10
.param dagetime='(10-0)'

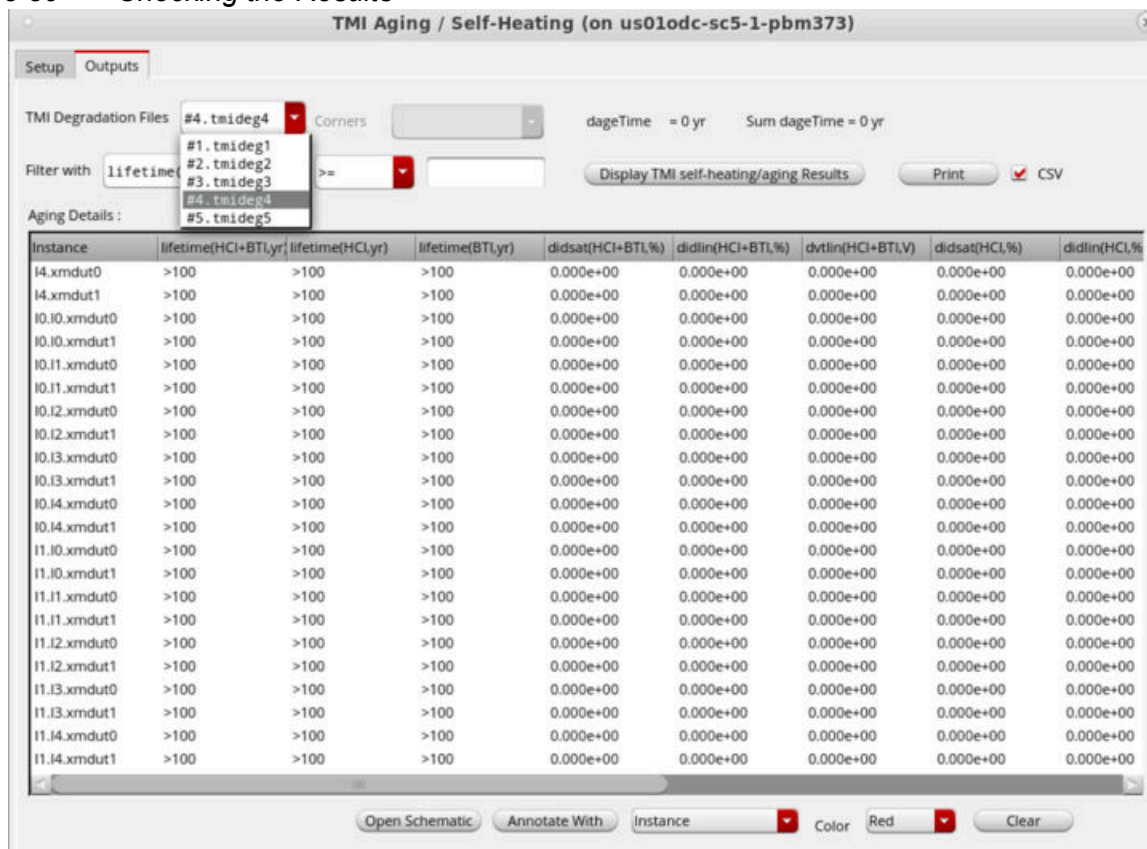
* Multi Stress & Aging Index = 2
.alter $Stress Simulation
.option tmiAge=1
.param dagetime=0
.temp 30
.param voltage=3
* Degraded Simulation
.alter $AgingTime=5
.param dagetime='(5-0)'
.temp 27
.param voltage=0.72

* Multi Stress & Aging Index = 3
.alter $Stress Simulation
.option tmiAge=1
.param dagetime=0
.temp 50
.param voltage=1.5
* Degraded Simulation
.alter $AgingTime=5
.param dagetime='(5-0)'
.temp 27
.param voltage=0.72
simulator lang=spectre
```

6. Check the results.
 - a. Open the **TMI Aging / Self-Heating** form.
 - b. Click the **Outputs** tab.

You can check the simulation results for all instances here.

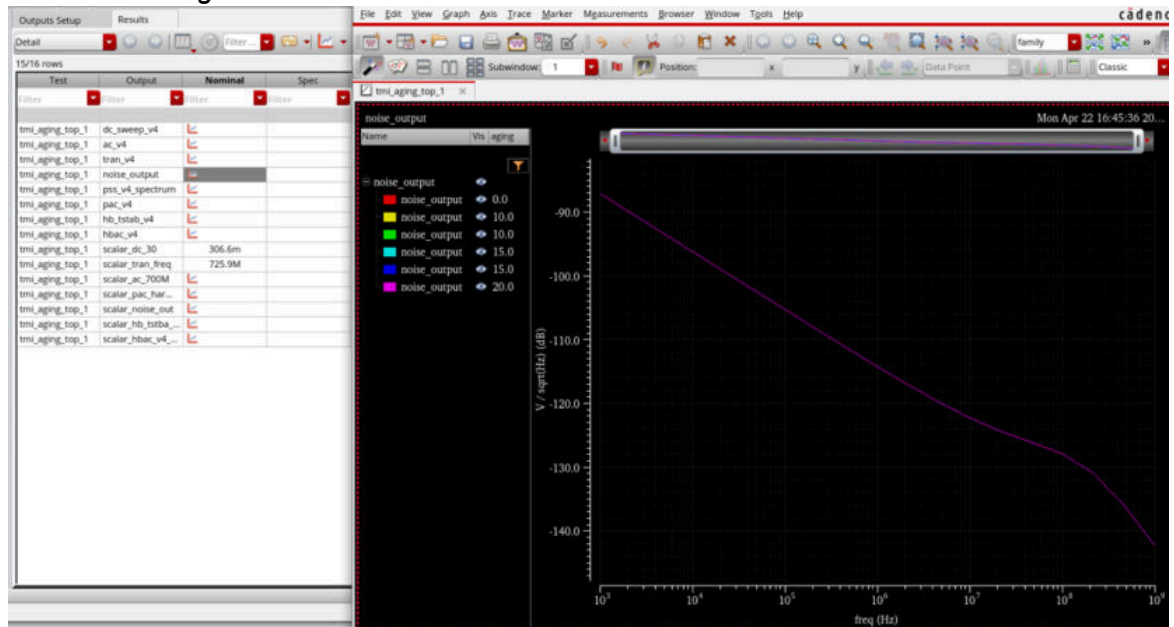
Figure 39 Checking the Results



7. Plot the aging waveforms.

- Open the **Results** table.
- Click the waveform icon to plot the aging waveforms.

Figure 40 Plotting the Waveforms

**Note:**

- The job distribution is controlled by ADE. For example, all settings in **Job Policy Setup** form are valid for PrimeSim SPICE Reliability Analysis.
- Both linear and RF analyses are supported in PrimeSim Reliability analysis.

7

Troubleshooting

This chapter discusses some common issues encountered while running ADE Explorer and how to troubleshoot them.

Assembler Error When Adding New Analyses

When trying to add a new analyses from Assembler, an error comes up and the **Choosing Analyses Form** fails to pop out .

It is a third-party bug. It is fixed in Virtuoso Studio. To resolve this error, update to Virtuoso Studio.

Cannot Run Simulation in Explorer or Assembler Due to XVFB

Clicking **Run Simulation** button on Explorer or Assembler shows an error message related to XVFB.

Figure 41 XVFB Error Message



To resolve this error, check whether the **CDF_XVFB_PATH** is set to the right installation path of XVFB. For more information, see the section *Installing the PrimeSim ADE Integration*.

Cannot Use LSCS Mode With PrimeSim

In Virtuoso Studio, Virtuoso starts to open the LSCS job distribution mode to third-party simulators.

To resolve this error, update to Virtuoso Studio if using LSCS with PrimeSim. Make sure to add this line in `working_dir/.cdsinit`:

```
envSetVal("maestro.lscs" "useLSCSFor3rdPartySimulators" 'boolean t)
```

Note:

If the PrimeSim is V-2023.12-SP1 or newer versions, you do not need to add this environment variable.

LSCS Performance Issue

When using LSCS mode in Job Setup, ADE may have performance issue.

The issue is fixed in PrimeSim V-2023.12-SP1-1. For older PrimeSim versions, the workaround is as follows:

1. Launch Virtuoso.
2. In the working directory, open the file `./vdsinit` file.
3. At the bottom of `./vdsinit` file, add the following code:

```
defmethod(relxPostSimulation ((session PrimeSim_SPICE_session)) t)
```

4. Run the simulation.

Note:

`./vdsinit` is deleted after existing Virtuoso. So, you need to perform above steps again on launching Virtuoso next time.

Assembler Does Not Plot Automatically in VIVA After Simulation

By default, ADE Explorer plots signals in VIVA automatically after simulation, but ADE Assembler does not.

To change the default behavior (not recommended):

1. In the Assembler menu, select **Options > Plotting/Printing**.
2. Change the **Plotting Option** from **None** to **Auto**.

OASIS Licenses-Related Issues

ADE needs OASIS licenses to run. This is also true for PrimeSim-ADE interface.

If there is an error or warning related to OASIS licenses, check whether there are sufficient OASIS licenses.

HB and HB Small Signal Analyses: Accuracy Difference

PrimeSim SPICE versus third-party simulator may have accuracy difference between HB and HB small signal analysis.

To resolve the issue:

1. Check whether the HB first tone **Number of Harms** is set to **auto**. PrimeSim SPICE and third-party simulator may use different number of harmonics.
2. To compare the signal analysis fairly, set the **Number of Harms** to the same number in both simulators.